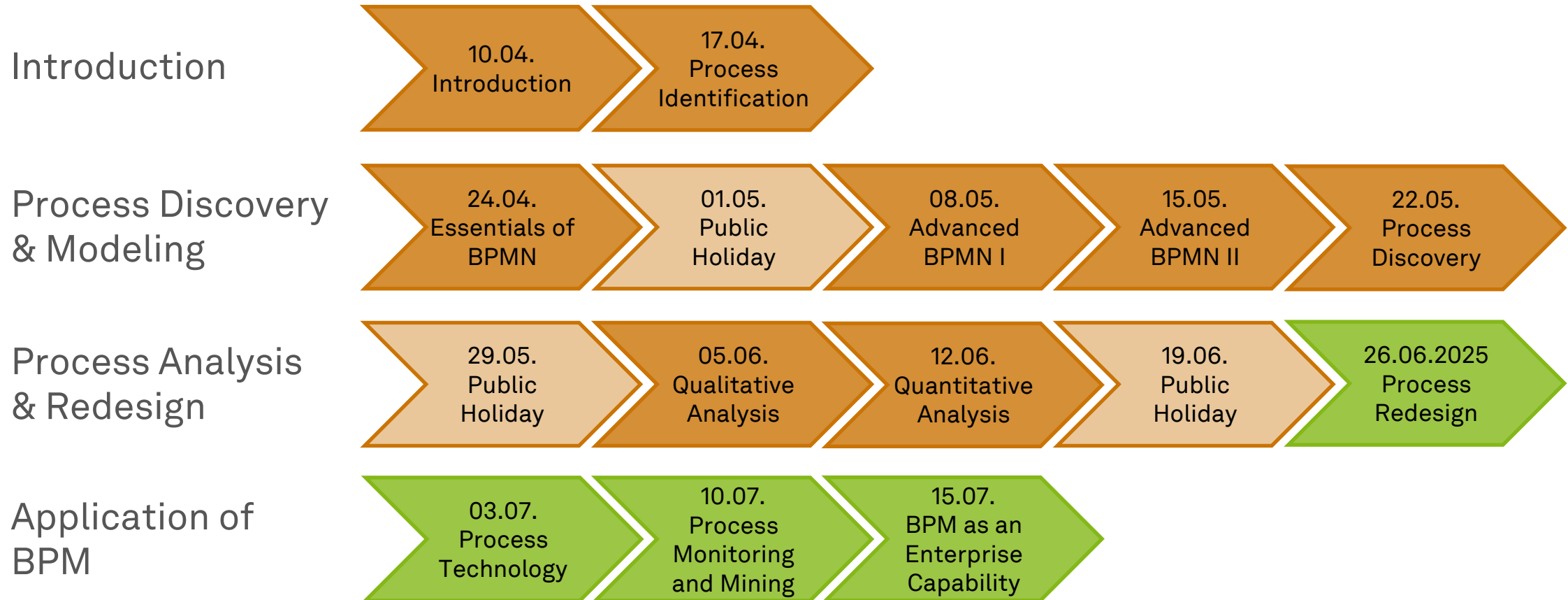


Business Process Management

Process Redesign



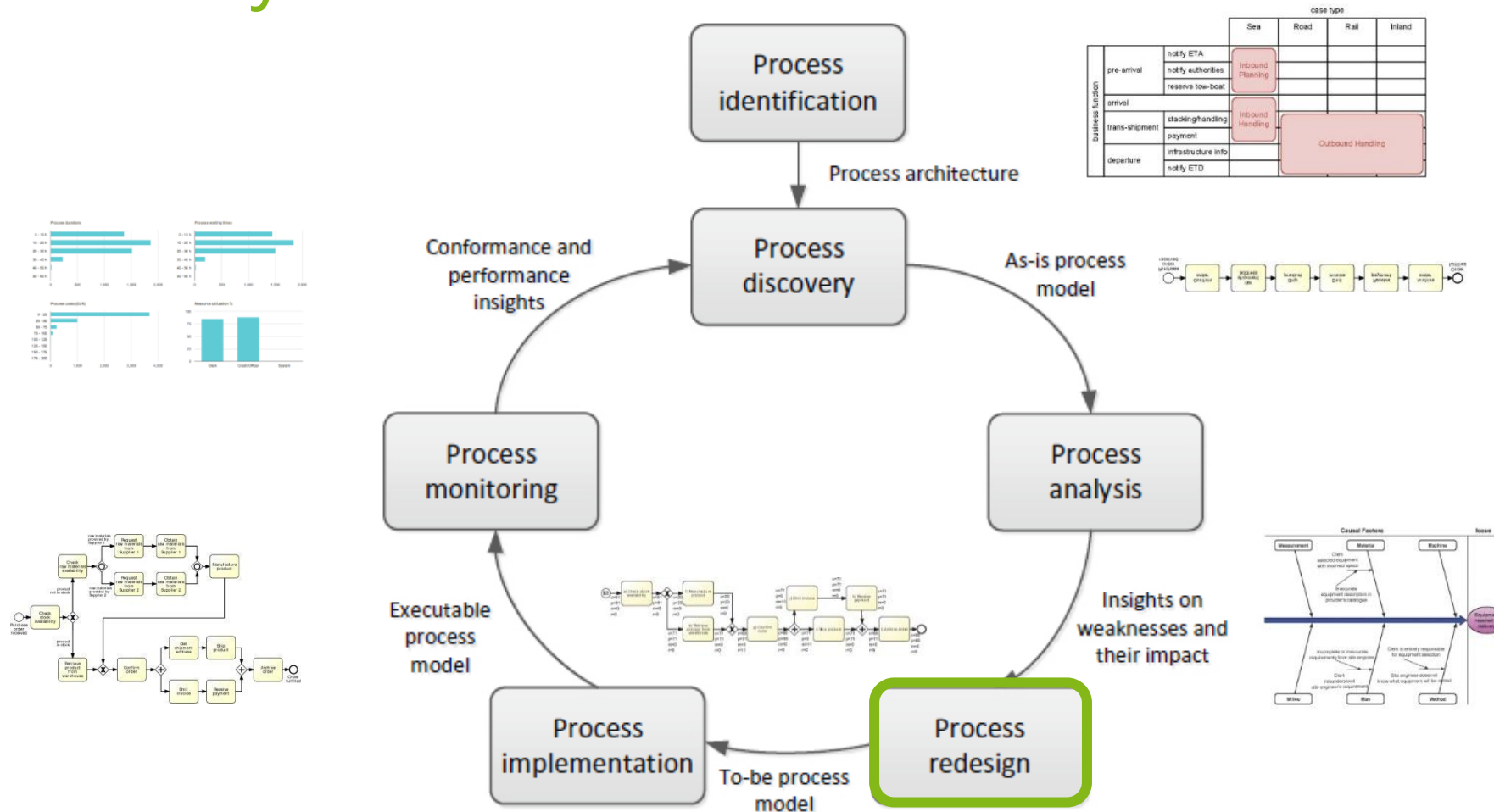
Roadmap



Learning Goals of Today

- Understand the essence of process redesign
- Understand the process redesign orbit
- Know and be able to use redesign techniques

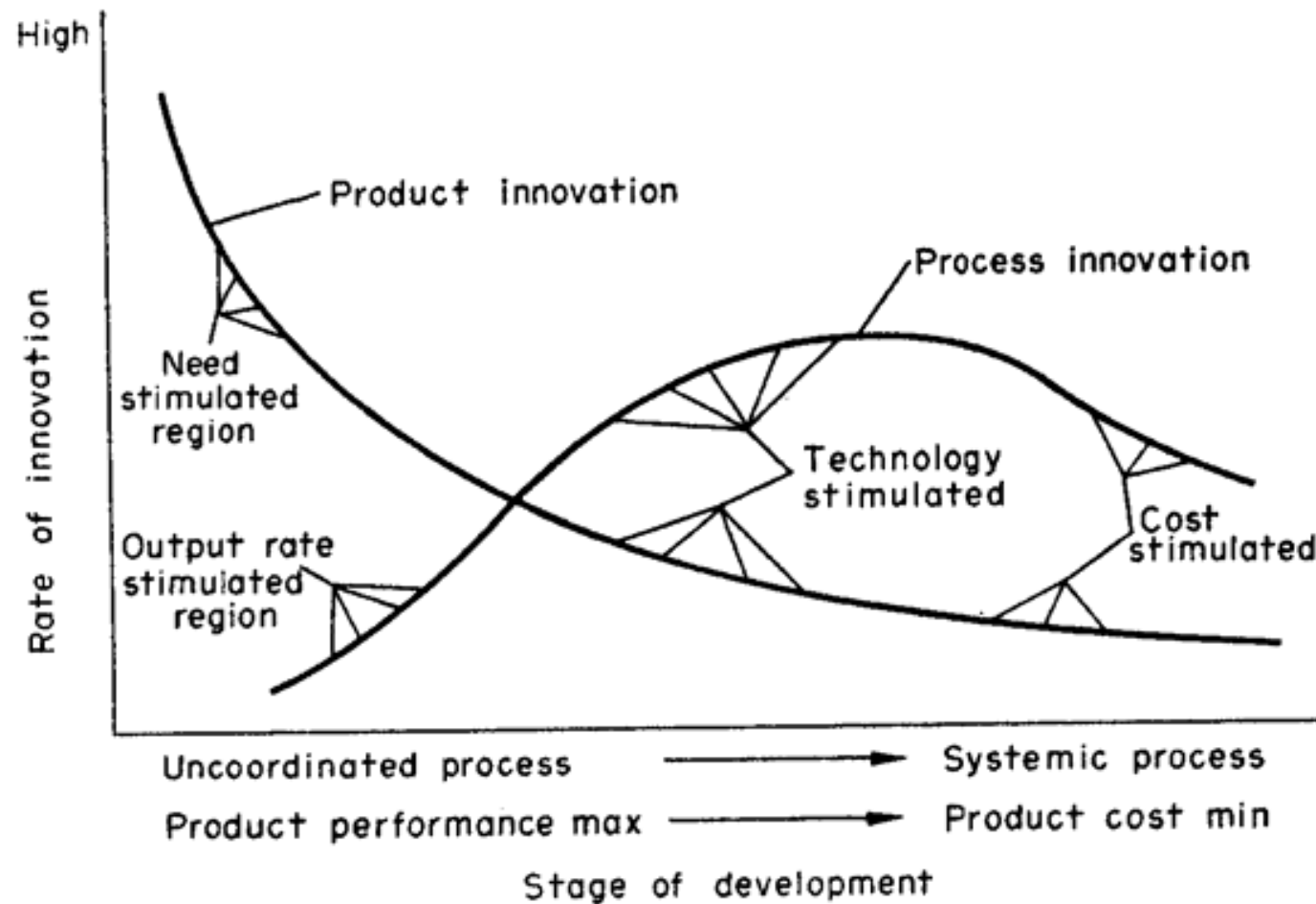
The BPM Lifecycle



Process Redesign



Product and Process Innovation



Motives of Process Redesign

- Process innovation
 - Complements product innovation
 - **Positive and proactive motive to process redesign**
- Organizational entropy
 - All processes evolve over time
 - Processes become more complex, and performance deteriorates
 - **Negative and reactive motive to process redesign**
- Process redesign does not only focus on existing but also on new processes

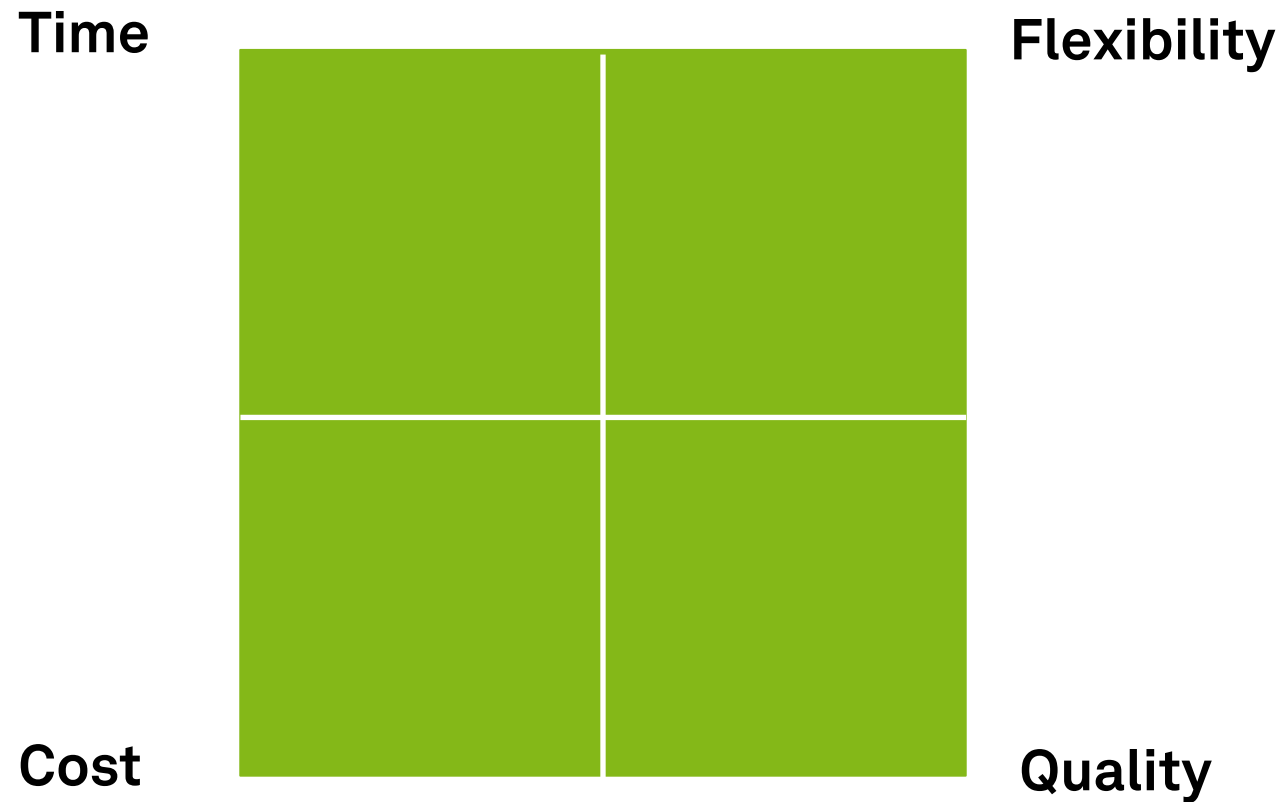
Aspects of Process Redesign

- Internal or external **customers**
- Business process **operation** view
 - Number and nature of activities
- Business process **behavior** view
 - Ordering and scheduling of activities
- **Organization** and the participants
 - Organization structure (groups, roles, etc.) and organization population (individuals)
- **Information** used and created
- **Technology** used
- **External environment**

Problems with Redesign

- **Complex** structures
 - not transparent: badly controllable, fault intolerant
- System **proliferation**
 - overlap, difficult information exchange, uncomfortable, bad maintainability
- **Inflexible** structure
 - change = pulling a house of cards
- Process was perhaps **once designed**, but **organic evolution degraded it** for years...

The Devil's Quadrangle



The Devil's Quadrangle – Example



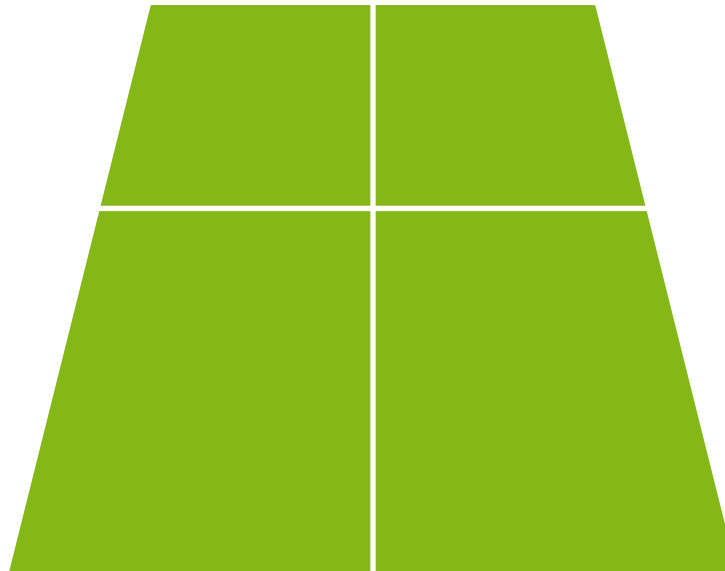
The Devil's Quadrangle – Example

Time

Flexibility

Cost

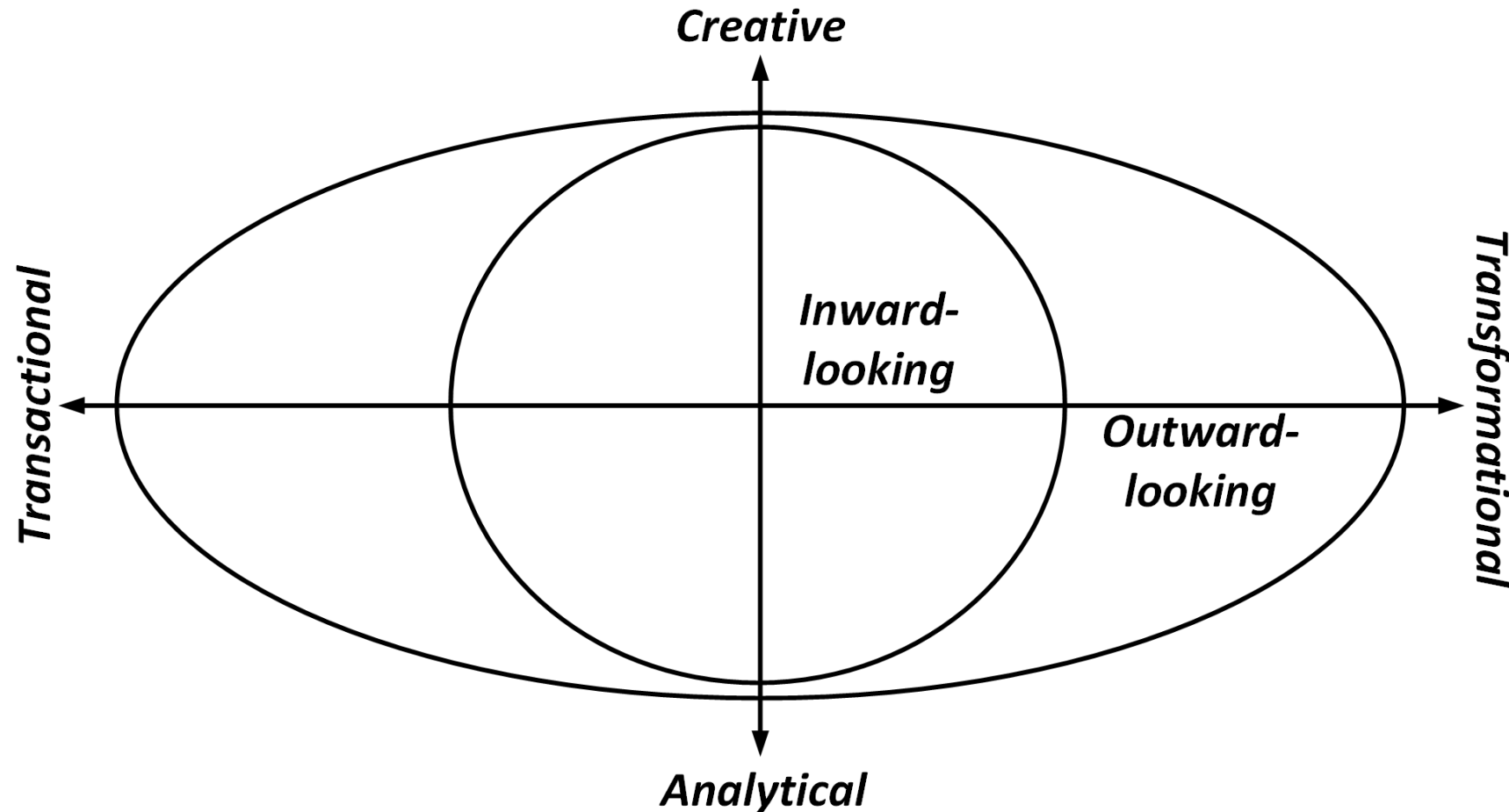
Quality



How to Redesign?

- Ambition and Situation?
 - **Revolutionary / Exploration**
 - Fundamental rethinking and radical redesign
 - **Evolutionary / Exploitation**
 - Incremental adaptation of process
 - Starting from
 - Clean slate
 - Existing process
 - Reference model/ blueprint
- How to redesign?
 - **Method, technique, tool**
- **There is no silver bullet!**

Process Redesign Orbit



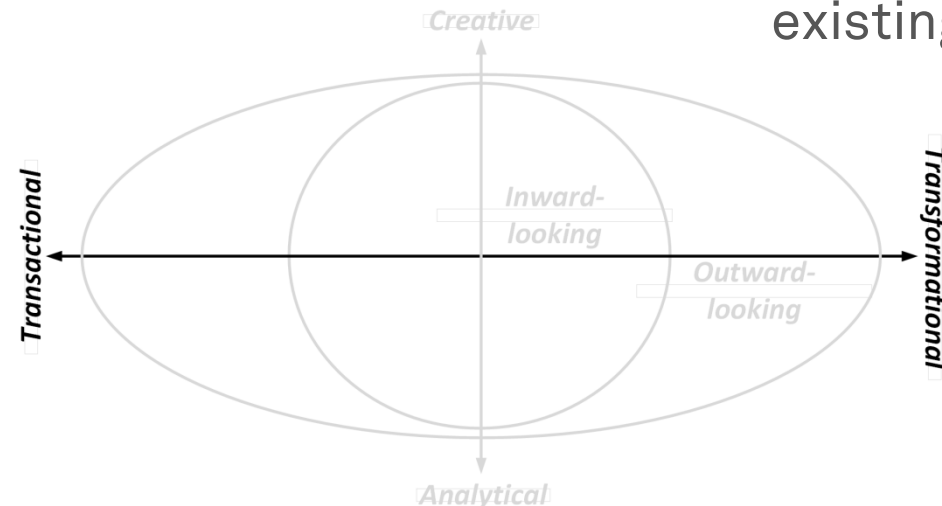
Process Redesign Orbit – Ambition

- **Transactional** Methods

- Seek to identify problems and resolve them **incrementally**, one step at a time
- Do not challenge the current process structure

- **Transformational** Methods

- Aim to achieve **breakthrough innovation**
- Put into question the fundamental assumptions and principles of the existing process structure



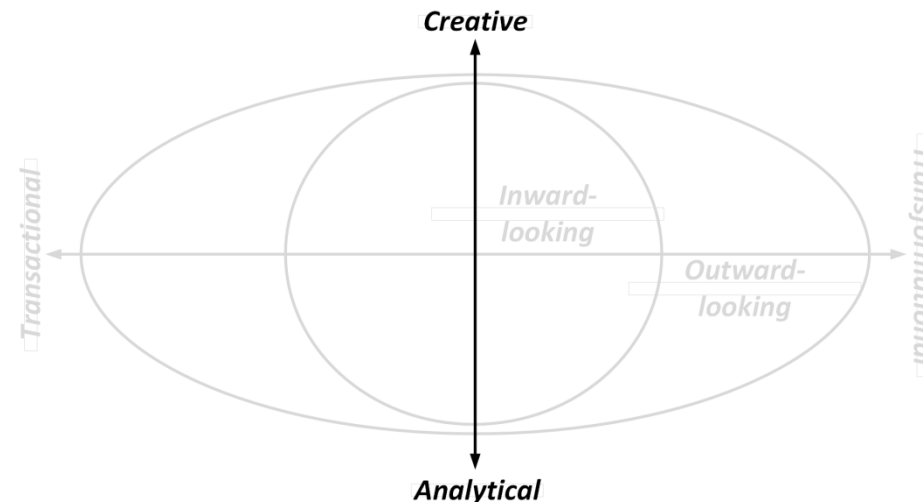
Process Redesign Orbit – Nature

- **Analytical** Methods

- Tend to have a strong mathematical and quantitative focus
- Embrace **tools** and **technology**

- **Creative** Methods

- Rely on (human) creativity and ingenuity
- Embrace **group dynamics**



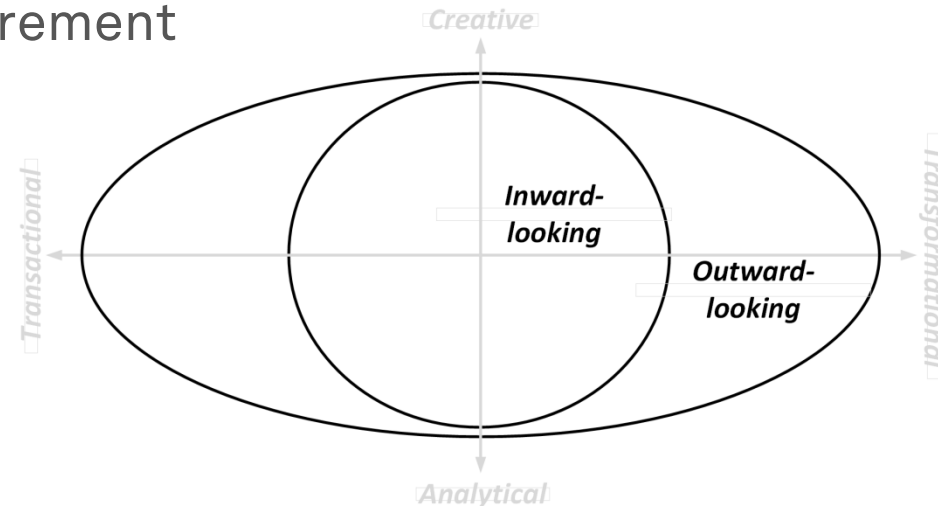
Process Redesign Orbit – Perspective

- **Inward-looking Methods**

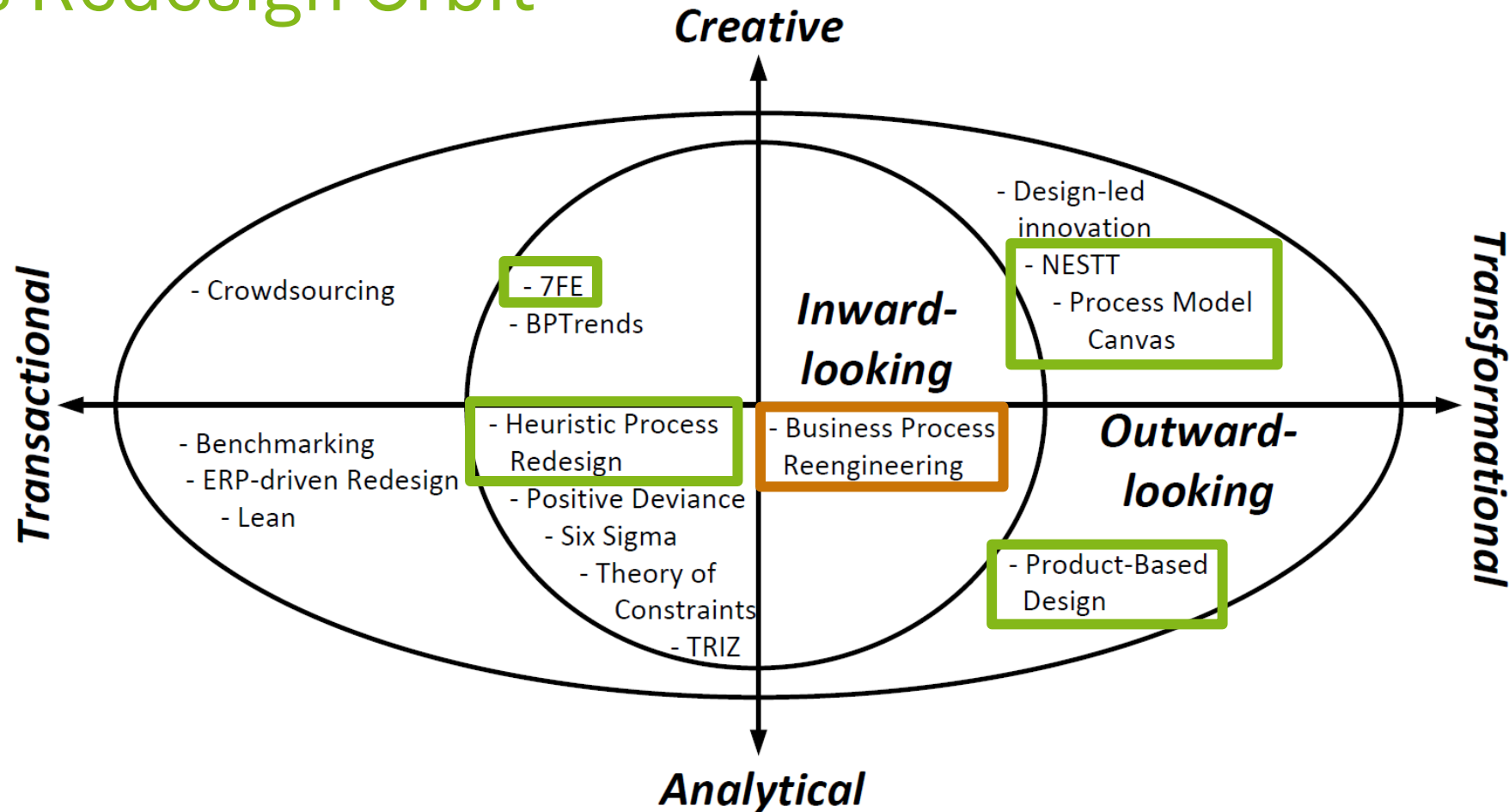
- Consider the process from the perspective of the **internal organization**
- Draw from objectives and performance measurement

- **Outward-looking Methods**

- Consider the process from an **outsider's perspective**
- Are driven by external opportunities and developments



Process Redesign Orbit

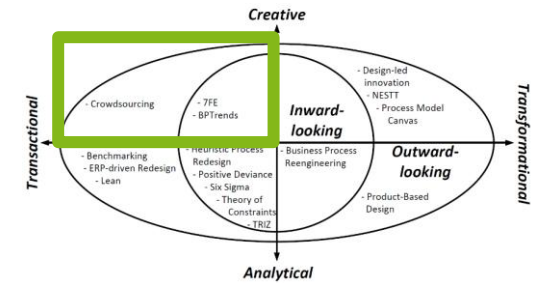


Creative Transactional Methods



Creative Transactional Methods

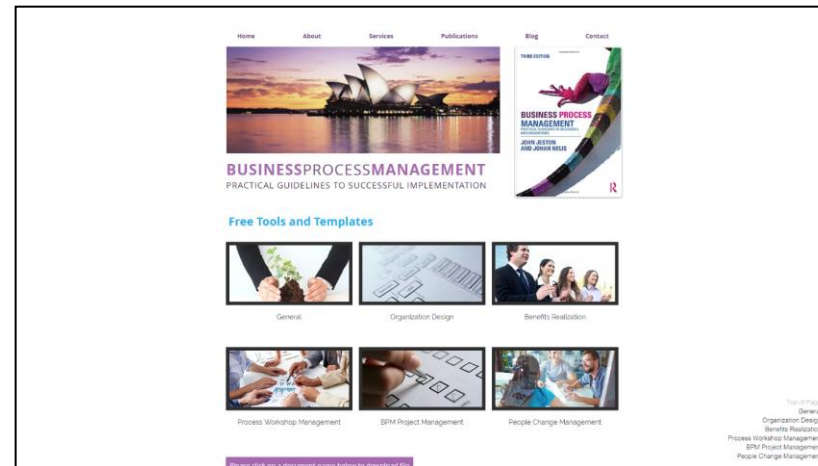
- Unleashing the creativity of people
- Inward-looking
 - 7FE
 - BPTrends
- Emphasis on engaging professionals that already play a role within the business process



- Outward-looking
 - Crowdsourcing
- Tapping into the skills and knowledge of people outside their organizational borders

7FE

- Framework for BPM projects or even BPM programs
- Consists of several phases
 - **Innovate** phase deals with the technical challenge of process redesign
- Method suggests conducting **workshops**



Stages of the Innovate Phase

- **Prepare**
 - Collect inputs for workshops
 - strategy link, goals and measures, constraints, timeframe
- **Generate**
 - Generate ideas with facilitator, converge to consensus
 - Redesign from **customer perspective**
 - Identify the **customer journey**
 - Identify (issues at) customer **touch points**
 - Use **mystery shopper**
 - Guide through **problematic elements** and **generic solutions**
 - Determine scenarios
- **Validate**
 - Test effectiveness and feasibility (with simulation)

7FE

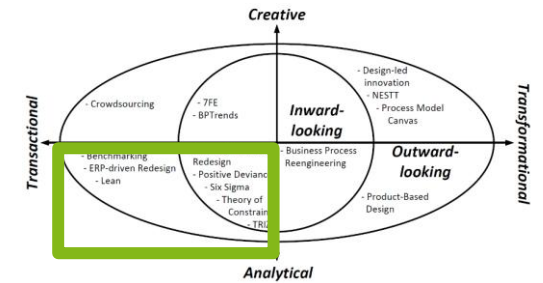
- **Facilitators...**
 - ...need to ask lots of “what if” and “why this” questions
 - ...should not accept what he or she is told (the first time)
 - ...must look for the second ‘right’ answer
 - ...will regularly change the question and come it at from a different direction
 - ...shall challenge the rules of the process
 - ...need to rely on intuition
- ...need vast experience

Analytical Transactional Methods



Analytical Transactional Methods

- Employing all kinds of tools, involving statistics
- strongly rationalized
- Inward-looking
 - Six Sigma
 - Theory of Constraints
 - TRIZ
 - Positive Deviance
 - **Heuristic Process Redesign**
- Focus on existing process in an organization as a starting point



- Outward-looking
 - Benchmarking
 - ERP-driven Redesign
 - Lean
- Take an external perspective on process redesign

Heuristic Process Redesign Procedure

- Uses a list of 29 redesign heuristics to determine improvement actions
- **Initiate**
 - Set up project
 - Create an understanding of the existing situation (**as-is**)
 - Set the **performance goals** for the redesign project
- **Design**
 - **Identify suitable heuristic** and **use heuristics** to determine potential improvement
 - **Generate scenarios** (to-be) as design alternatives
- **Evaluate**
 - Qualitative and/or quantitative **analysis of** redesign **scenarios**
- Execute in highly **iterative and overlapping** ways

Examples of Time-based Redesign Heuristics

Parallelism

- *“Put activities in parallel”*
- Activities are often ordered in a **strictly sequential** way
- Some activities may well be carried out in an **arbitrary order** or even **simultaneously**
- With less restrictive choice on the order, a business process can be carried out **faster**

Case-based work

- *“Remove batch-processing and periodic activities”*
- **Sources of delays**
 - batch that is only processed once all its items are available
 - periodic triggers
- Removing such constraints is in general a good way to **speed up** a process

Examples of Cost-based Redesign Heuristics

Activity elimination

- “*Eliminate unnecessary activities*”
- Organizational entropy entails that processes **build up unnecessary steps** over time
- Removing unnecessary activities is an effective way to **reduce the cost** of handling a case

Empower

- “*Give workers decision-making authority*”
- Often other roles people **authorize** the outcomes of activities
- Empowerment to take decisions autonomously may remove some work of middle managers and **reduce cost**

Examples of Quality-based Redesign Heuristics

Triage

- “Split an activity into alternative versions”
- Alternative versions of an activity can better deal with **variants of cases**
 - Deal with a sub-category of cases
 - Exploit characteristics of a resource
- By aligning work more specifically to a particular case, the **quality of work improves**

Case assignment

- “Let participants perform as many steps as possible”
- A resource becomes acquainted with the case
- Make this participant the **preferred resource** for any work in this case
- This knowledge can be leveraged to deliver a **high standard of work**

Examples of Flexibility-based Redesign Heuristics

Flexible assignment

- “Keep generic participants free for as long as possible”
- Assign cases to the most specialized resource first
- Maximizes the **likelihood to commit** the free, more general participant to another work item
- Process **stays flexible** with respect to assigning work

(Virtualized) Centralization

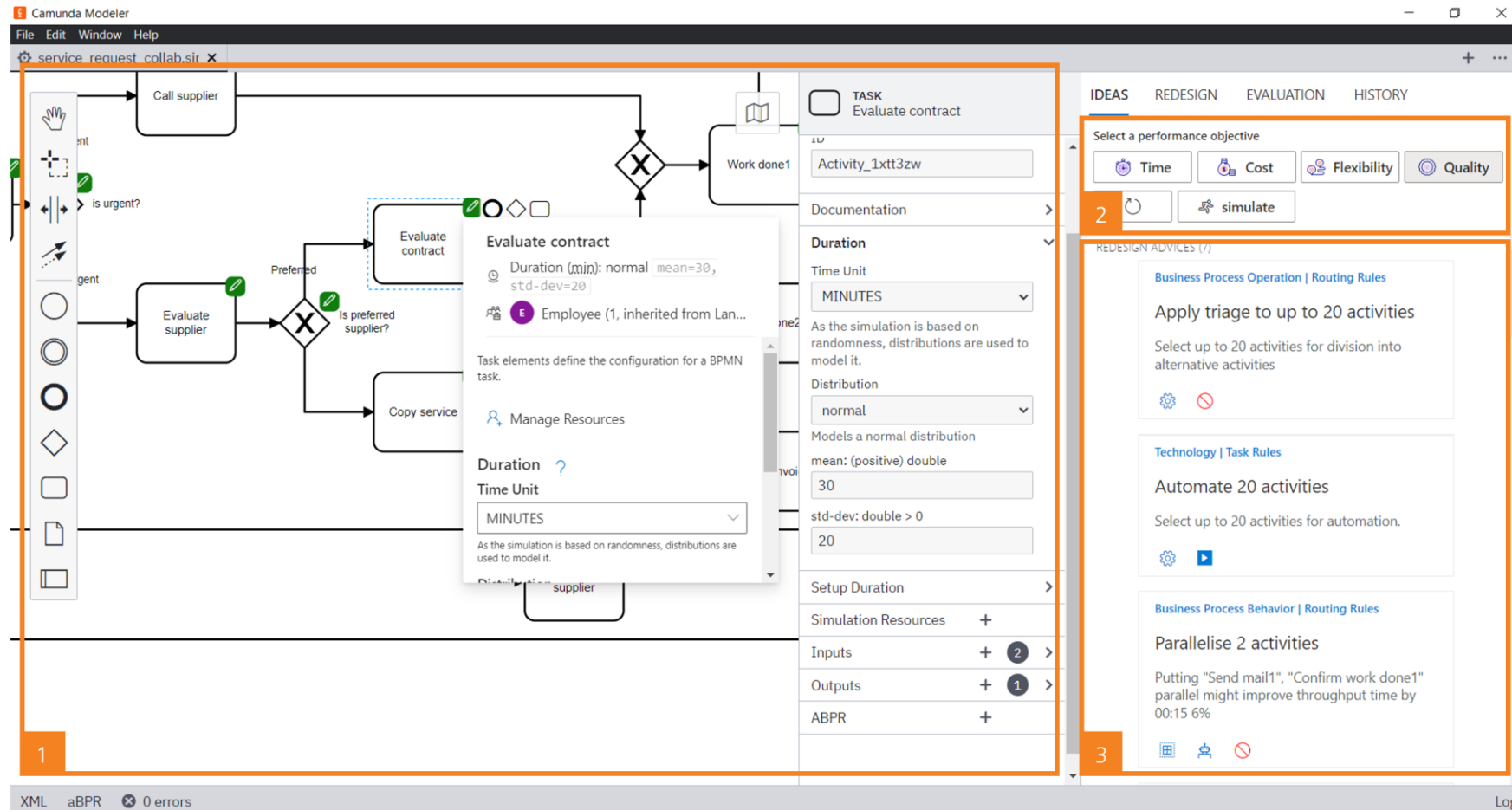
- “Let geographically dispersed resources act as if they are centralized”
- Exploit the benefits of a BPMS
- Geographic allocation becomes irrelevant
- Resources can be **committed more flexibly**

Examples of Redesign Heuristics

Dimension	Heuristic	Time	Cost	Quality	Flex.
Time	Parallelism	+			
	Case-based work	+			
Cost	Activity elimination	+	+		
	Empower	+	+		+
Quality	Triage			+	
	Case assignment	+		+	
Flexibility	Flexible assignment	+		+	+
	Centralization	+			+

Redesign
Heuristics can
also have
negative effects
(see Devil's
Quadrangle)

A Tool for Assisted Business Process Redesign

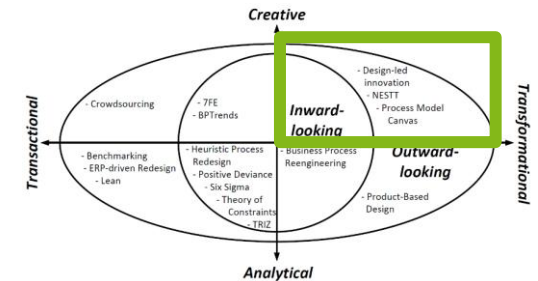


Creative Transformational Methods



Creative Transformational Methods

- Unleashing the creativity of people
- Inward-looking
 - *(True transformations hardly emanate from reasoning from an internal perspective only)*
- Emphasis on engaging professionals that already play a role within the business process
- Outward-looking
 - BPM Billboard
 - Design-led innovation
 - **NESTT**
 - **Process Model Canvas**
- Tapping into the skills and knowledge of people outside their organizational borders



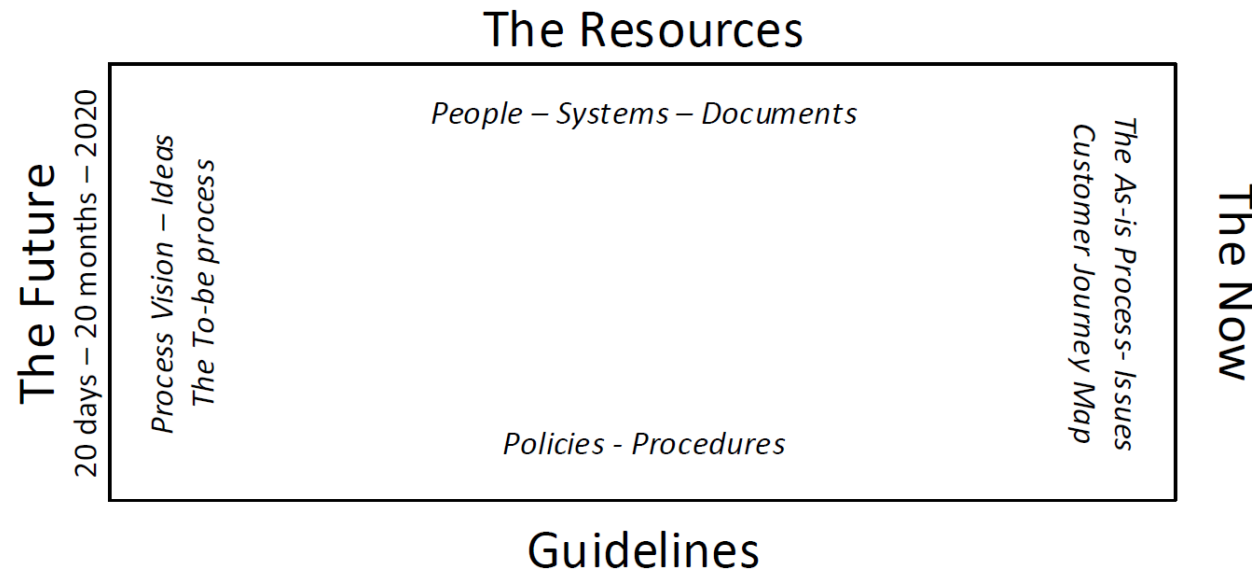
NESTT

- **Stages**

- Navigate, Expand, Strengthen, Tune/Take-off

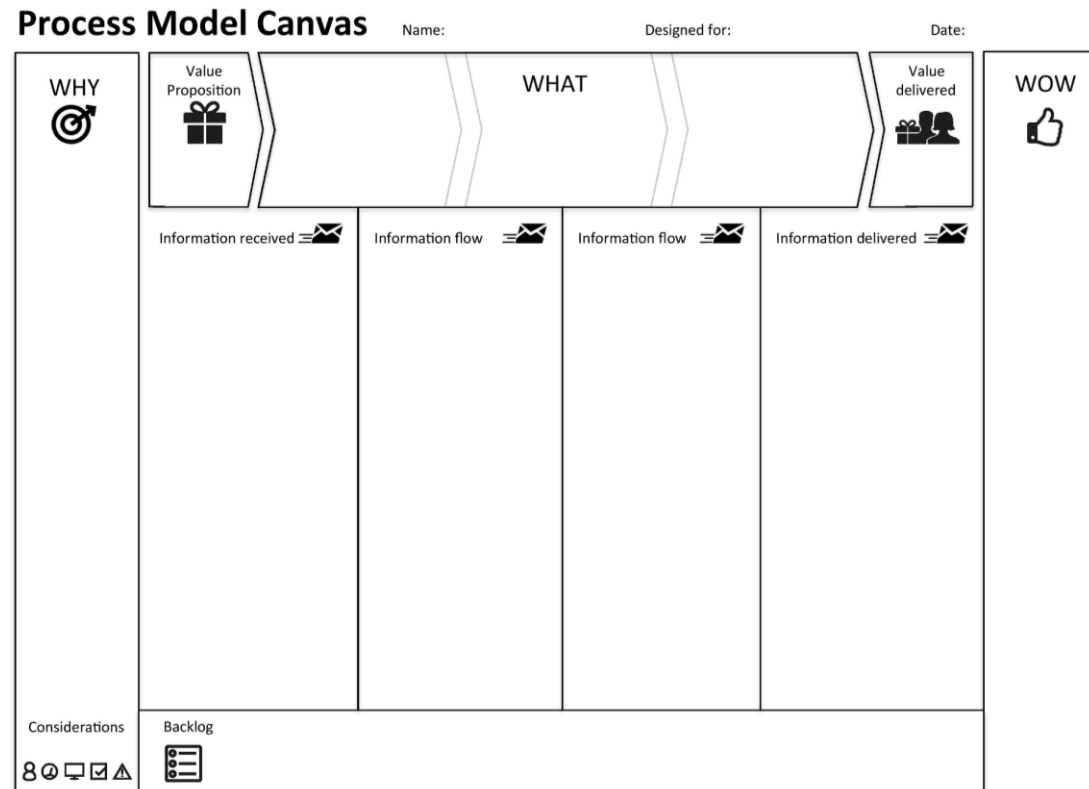
- **Workshop setting**

- 8 to 10 people use the walls and the floor of the room to visualize and address different viewpoints



Process Model Canvas

- Allows firms to reason about the **value proposition** behind their processes
- Questions
 - Why?
 - What?
 - Wow!?



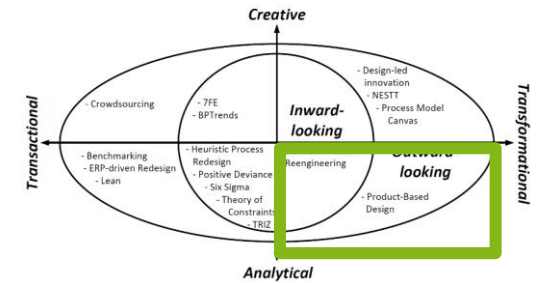
Dumas et al. (2018), processmodelcanvas.net (2022)

Analytical Transformational Methods



Analytical Transformational Methods

- Employing all kinds of tools, involving statistics
- strongly rationalized
- Inward-looking
 - Business process reengineering
- Focus on existing process in an organization as a starting point
- Outward-looking
 - Product-based design
- Take an external perspective on process redesign



Business Process Reengineering

Caveat: Has practically
been superseded with
more transactional
approaches



- **Clean slate approach**
 - Successful organizations do not rely on piecemeal improvement
 - Break away from ingrained patterns, integrate cross-functionally
- **Information technology**
 - Successful organizations do not only automate, they innovate
- **Product (or service) is central concept rather than process**
- **Principles**
 - Make sure information is fresh from the source
 - Integrate information capture and processing with the real work
 - Let those that have interest in the output drive the change, not only participate
 - Put decision points at the place where work is performed

Dumas et al. (2018), Hammer (1990)

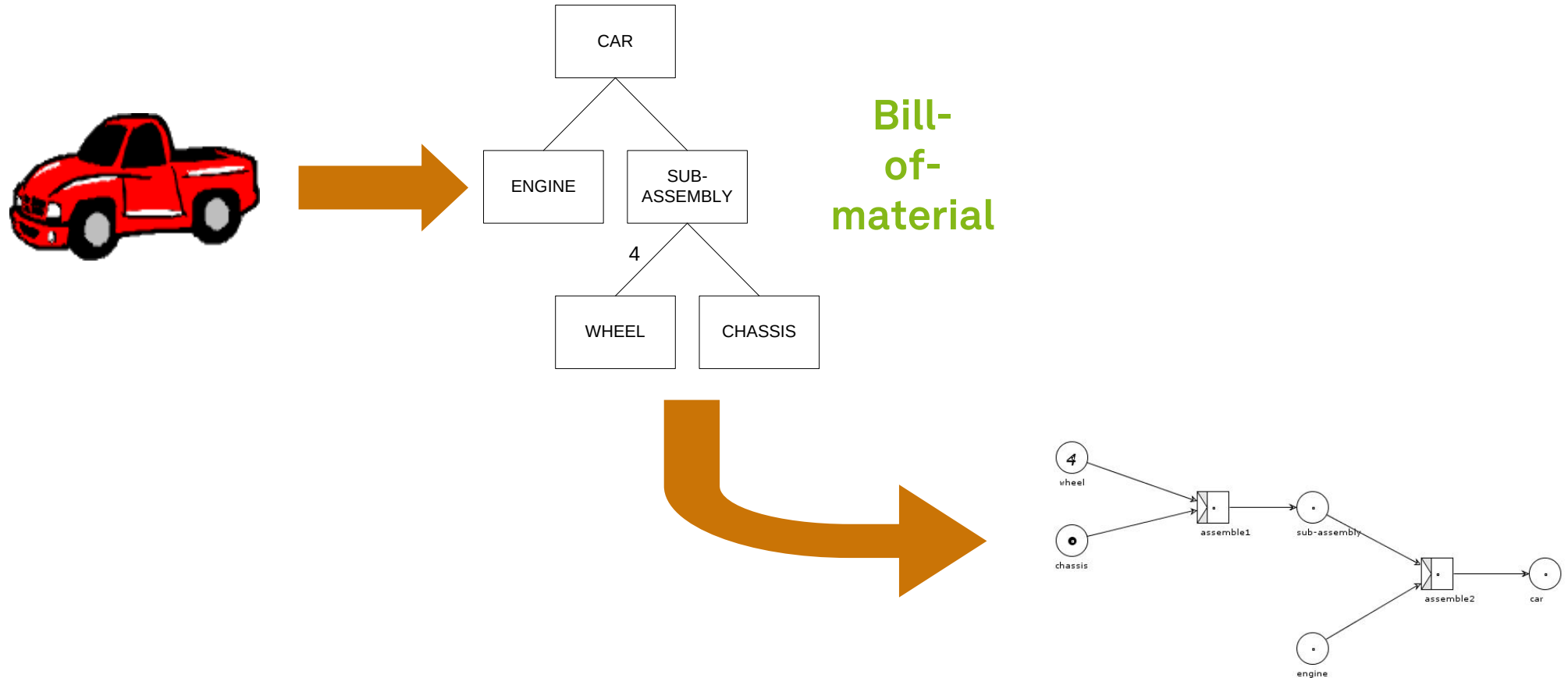
Product-based Design

- Idea to **ignore the existing process** and purely **consider the features of the product or service**
- Develop the **leanest, most performative** process possible
- Developed to design processes that produce informational products
- **Product data model** (bill-of-material) is the main vehicle to determine the best process structure to create and deliver that product

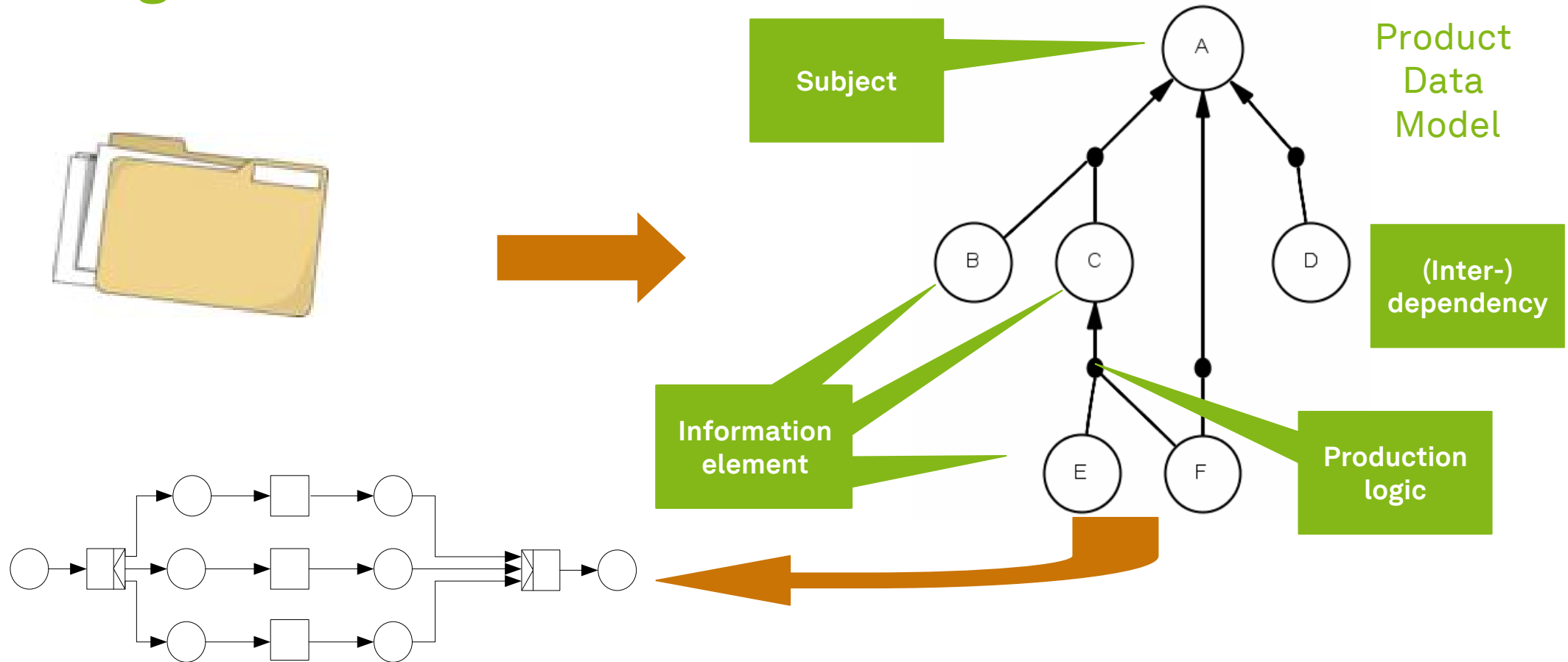
Product-based Design Procedure

- **Scoping**
 - Select **subject** of redesign
 - Identify performance targets and limitations
- **Analysis**
 - Create and study product data model
- **Design**
 - Derivate (multiple) designs using elements and operations
- **Evaluation**
 - Verify, validate, and select
- Execute in an **iterative** ways

Manufacturing Process Model



Exploiting the Product Data Model



Scoping – Information Elements

- Information elements should be **adequate in size**
 - Information element is too large if different parts of it are used in different production rules
 - Information element should not contain irrelevant information
- Information elements should **not be** necessarily **associated with** their **physical manifestation**
 - Avoid information elements like “computer screen output”
- Information elements may be **atomic** or **composite**
 - The type of a composite information element is a composed type

Scoping – Production Logic

- The full **specification** is a direct **validation** on the inputs of the involved production rule
 - Forgotten inputs or bad data types can be detected
- An **analysis** of the **production logic** is **relevant for** the estimation of **performance** characteristics
 - Labor and computer cost, speed, accuracy, etc.
- Production logic of **algorithmic nature** can be **used** as a **functional specification** for the BPM system that can execute this production rule
- If the production logic is **not totally algorithmic**, it is **likely** that a **human operator** must execute it in practice
 - The production logic can be used to develop task instructions

Analysis

- **Source analysis**

- Identify the sources of all the leaf elements in the product data model
- Relevant points of comparison for each leaf are
 - cost of obtaining it, delivery speed of the information, availability of the specific information and reliability of the provided information

- **Production analysis**

- Estimate the involved cost, speed, quality, and flexibility
- Select the right set and the right execution order of production rules given a set of performance targets

- **Fraction analysis**

- Study the distribution of information element values
- Model the rule with the widest applicability first (reduce workload)

Design

- **Correct**
 - Activities may only be incorporated in a process model in a way such that the **dependencies** between the information elements in the product data model **are respected**
- **Complete**
 - A process design must **cover all the information elements** that have been identified in a product data model
- **Sound**
 - Verification and validation
- **Fast vs. efficient**
 - Speed vs. resource utilization
- **Secure**

Learning Goals of Today

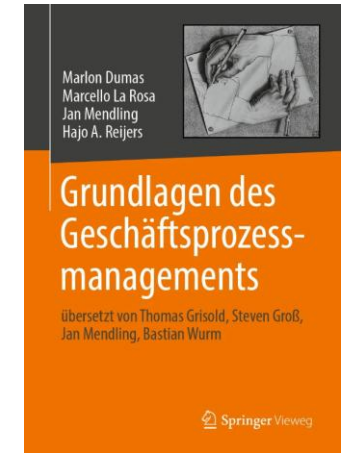
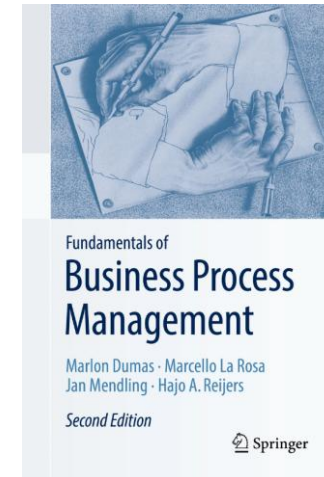
- Understand the essence of process redesign
- Understand the process redesign orbit
- Know and be able to use redesign techniques

Recap Questions

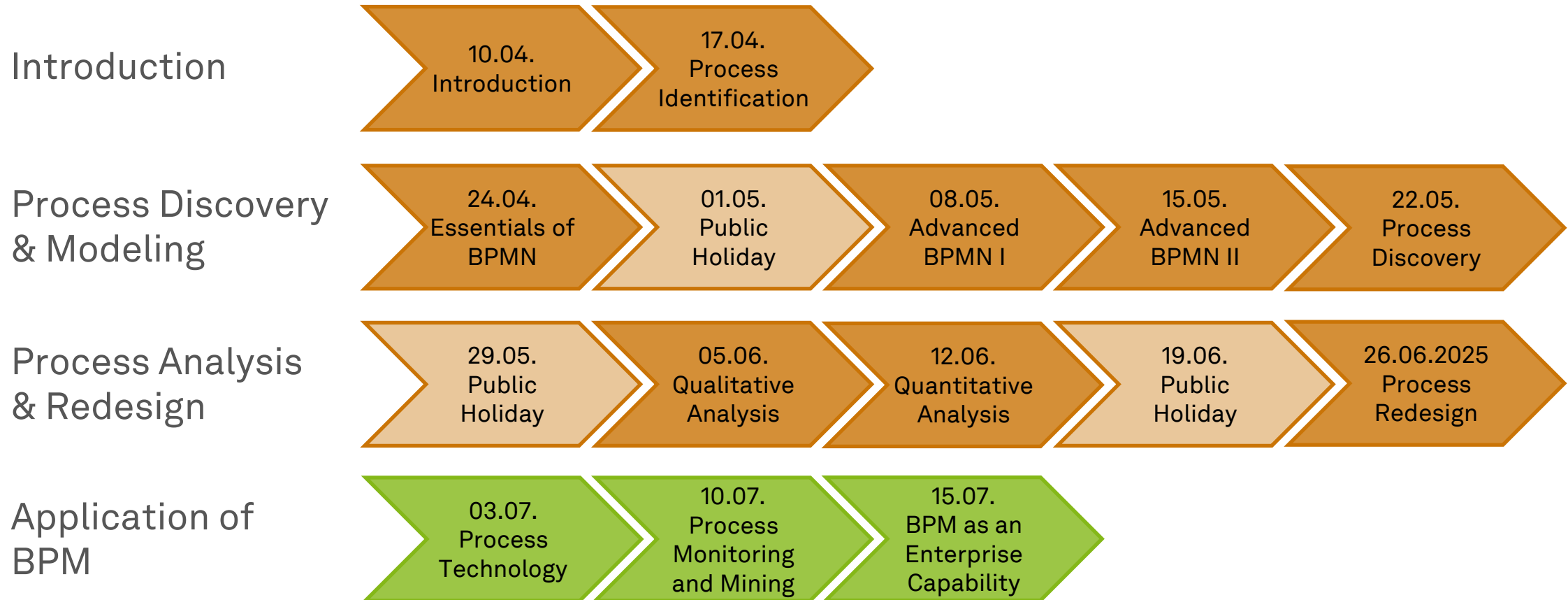
- What is the process redesign orbit and which three dimensions does it distinguish?
- How can heuristic process redesign be classified in the process redesign orbit and how does it work?
- What is product-based design? How does it work?

Literature

- **Section 8: Process Redesign**
- www.managementbyprocess.com
- www.processmodelcanvas.net



Roadmap



Any Questions about Process Redesign?





ec.cs.tu-dortmund.de

